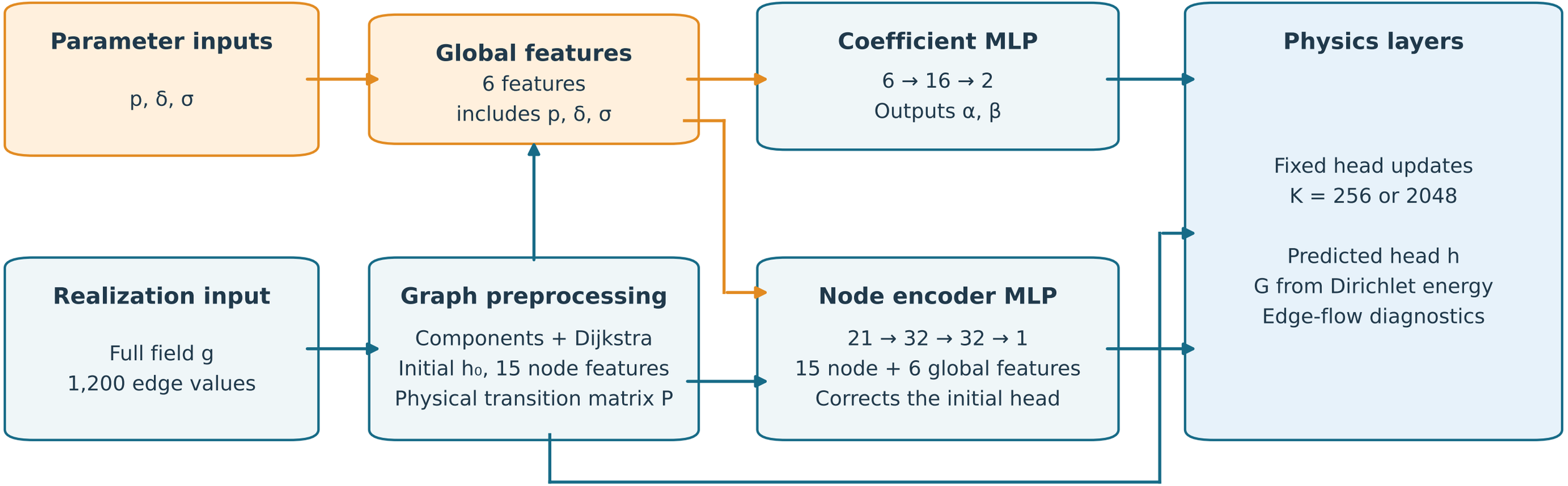
Parameter inputs
p, δ, σ
Global features
6 features
includes p, δ, σ
Coefficient MLP
6 → 16 → 2
Outputs α, β
Physics layers
Fixed head updates
K = 256 or 2048
Predicted head h
G from Dirichlet energy
Edge-flow diagnostics
Realization input
Full field g
1,200 edge values
Graph preprocessing
Components + Dijkstra
Initial h₀, 15 node features
Physical transition matrix P
Node encoder MLP
21 → 32 → 32 → 1
15 node + 6 global features
Corrects the initial head
Training: energy + conservation residual loss • 1,939 trainable parameters